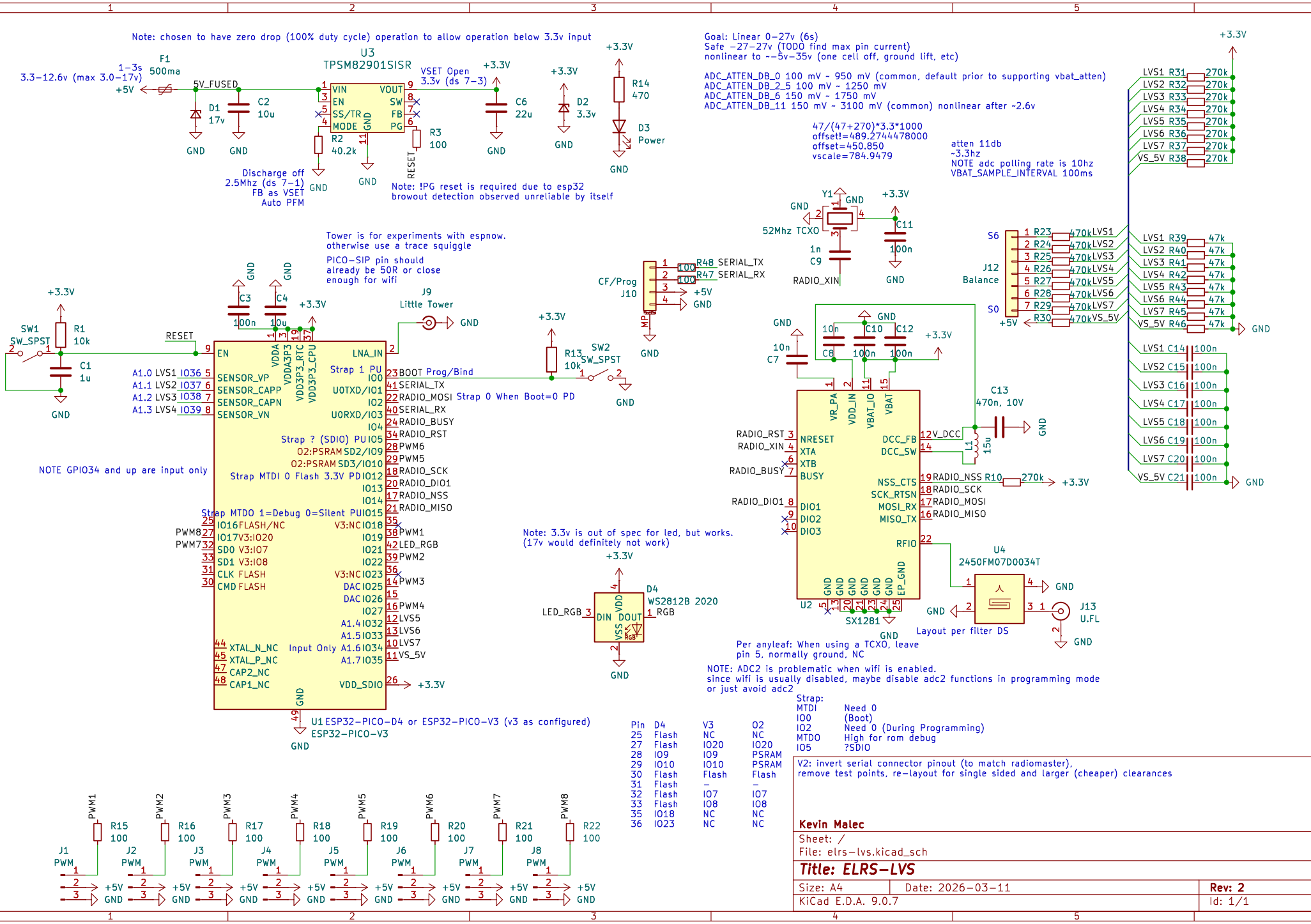
Note: chosen to have zero drop (100% duty cycle) operation to allow operation below 3.3v input
F1
500ma
1-3s
3.3-12.6v (max 3.0-17v)
5V_FUSED
U3
TPSM82901SISR
VSET Open
3.3v (ds 7-3)
D1
17v
C2
10u
C6
22u
D2
3.3v
R14
470
D3
Power
R2
40.2k
R3
100
Discharge off
2.5Mhz (ds 7-1)
FB as VSET
Auto PFM
Note: IPG reset is required due to esp32
browout detection observed unreliable by itself
Goal: Linear 0-27v (6s)
Safe -27-27v (TODO find max pin current)
nonlinear to ~-5v-35v (one cell off, ground lift, etc)
ADC_ATTEN_DB_0 100 mV ~ 950 mV (common, default prior to supporting vbat_atten)
ADC_ATTEN_DB_2_5 100 mV ~ 1250 mV
ADC_ATTEN_DB_6 150 mV ~ 1750 mV
ADC_ATTEN_DB_11 150 mV ~ 3100 mV (common) nonlinear after ~2.6v
47/(47+270)*3.3*1000
offset!=489.2744478000
offset=450.850
vscale=784.9479
atten 11db
~3.3hz
NOTE adc polling rate is 10hz
VBAT_SAMPLE_INTERVAL 100ms
Tower is for experiments with espnow.
otherwise use a trace squiggle
PICO-SIP pin should
already be 50R or close
enough for wifi
J9
Little Tower
CF/Prog
J10
SERIAL_TX
SERIAL_RX
52Mhz TCXO
Y1
C11
C9
RADIO_XIN
J12
Balance
S6
S0
SW1
SW_SPST
R1
10k
C1
1u
RESET
C3
C4
100n
SW2
R13
NOTE GPIO34 and up are input only
Strap MTDO 1=Debug 0=Silent PU
Strap MTDI 0 Flash 3.3V PD
Strap ? (SDIO) PU
Strap 1 PU
Strap 0 When Boot=0 PD
U1 ESP32-PICO-D4 or ESP32-PICO-V3 (v3 as configured)
ESP32-PICO-V3
Note: 3.3v is out of spec for led, but works.
(17v would definitely not work)
D4
WS2812B 2020
C10
C12
C7
C8
C13
470n, 10V
L1
15u
U4
2450FM07D0034T
J13
U.FL
SX1281
U2
Layout per filter DS
Per anyleaf: When using a TCXO, leave
pin 5, normally ground, NC
NOTE: ADC2 is problematic when wifi is enabled.
since wifi is usually disabled, maybe disable adc2 functions in programming mode
or just avoid adc2
Strap:
MTDI Need 0
IO0 (Boot)
IO2 Need 0 (During Programming)
MTDO High for rom debug
IO5 ?SDIO
Pin D4 V3 O2
25 Flash NC NC
27 Flash IO20 IO20
28 IO9 IO9 PSRAM
29 IO10 IO10 PSRAM
30 Flash Flash Flash
31 Flash - -
32 Flash IO7 IO7
33 Flash IO8 IO8
35 IO18 NC NC
36 IO23 NC NC
V2: invert serial connector pinout (to match radiomaster),
remove test points, re-layout for single sided and larger (cheaper) clearances
Kevin Malec
Sheet: /
File: elrs-lvs.kicad_sch
Title: ELRS-LVS
Size: A4
Date: 2026-03-11
Rev: 2
KiCad E.D.A. 9.0.7
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